

CSE200 LaTeX Tutorial II

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1 Text Formatting

Package note: bold, underline, and emphasis commands are built in. Text color needs the `xcolor` package. Colored table rows and cells need the `table` option for `xcolor`, as shown in the preamble.

The commands below are worth knowing for text formatting.

- `\textbf{...}` gives **bold text**.
- `\underline{...}` gives underlined text.
- `\emph{...}` gives *emphasized text*; its appearance may change depending on context.

Example sentence: A technical report should use **important terms**, precise labels, and *consistent emphasis* carefully.

Use `xcolor` when color has a clear purpose: emphasis, status, or readability. This sentence has **red text**, a **custom phrase**, and a **short highlighted note**.

Item	Result	Comment
Figure reference	Correct	Label placed after caption
Image size	Needs work	Width too large
Citation	Correct	Source appears in bibliography

Table 1: Color used to communicate status in a table.

2 Nested and Mixed Lists

Package note: basic `itemize` and `enumerate` lists are built in. Custom list labels in this section use the `enumitem` package.

Technical documents often combine numbered lists, bullets, custom symbols, and lettered sublists. The `enumitem` package gives reliable control over list labels.

1. System Setup

- Install software packages
- Configure hardware
- ! Verify connections

2. Simulation Steps

- (a) Initialize parameters

- (b) Run simulation
 - Monitor CPU usage
 - Record intermediate results
 - (c) Analyze outputs
3. Reporting
- Prepare charts
 - # Document observations
 - Summarize findings
 - (a) Key insights
 - (b) Recommendations

3 Image Files and the figure Float

Package note: image insertion needs `\usepackage{graphicx}`. The `figure` environment itself is built in.

Use `\includegraphics` to insert an image. Put it inside a `figure` environment when it needs a number, caption, label, and cross-reference.



Figure 1: Bangladesh observed by NASA's Terra satellite.

Figure ?? is referenced from the paragraph. The label comes after the caption so that LaTeX stores the correct figure number.

3.1 Image Size, Scale, and Rotation



Figure 2: The same figure can demonstrate width, scale, and angle options.

Prefer `width=...` with `\linewidth` for predictable layouts. Use `scale=...` when you want to resize by a factor, and `angle=...` for rotation.

4 Float Placement: `h`, `t`, `b`, `p`, `!`, and `H`

LaTeX floats are allowed to move. The placement option is a suggestion, not a command, except for `[H]` from the `float` package. Package note: `h`, `t`, `b`, `p`, and `!` are built in; `H` requires `\usepackage{float}`.

Option	Meaning	
<code>h</code>	here	
<code>t</code>	top	
<code>b</code>	bottom	
<code>p</code>	page	
<code>!</code>	override	Relax some internal restriction
<code>H</code>	exactly here	Requires <code>float</code> ; useful for demos and

Table 2: Common float placement options.

4.1 Placement Examples

There are no standard capital variants such as `T` or `B` for ordinary article floats. The important capital option for this lesson is `H`, provided by the `float` package.

5 Wrapped Figures

Package note: wrapped figures need `\usepackage{wrapfig}`.

A wrapped figure lets text flow beside an image. This is useful when a small figure belongs to the same paragraph as its explanation. Keep enough text after the wrapped figure; otherwise, the next paragraph may begin too close to the float. Also avoid making wrapped images too wide. A width between `0.25\linewidth` and `0.40\linewidth` is

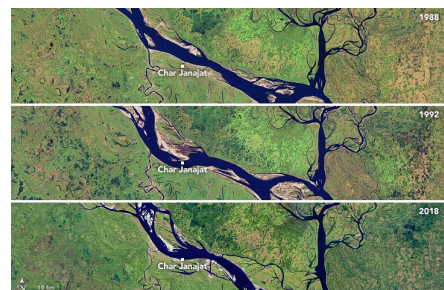


Figure 6: Wrapped image.



Figure 3: A top-position float using `[t]`.

often a good starting point. Figure ?? shows a compact layout that is useful when an image and its explanation should stay visually close.

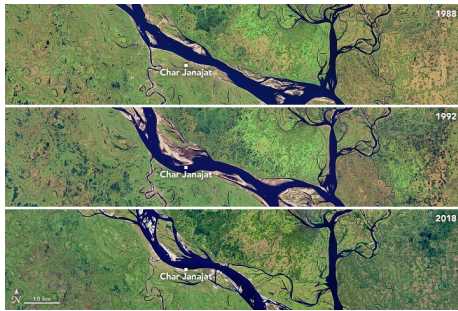


Figure 4: A bottom-position float using `[b]`.

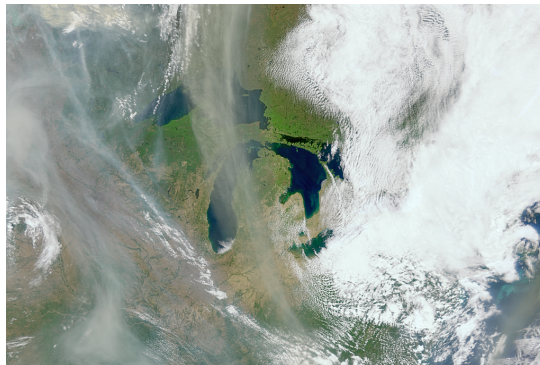
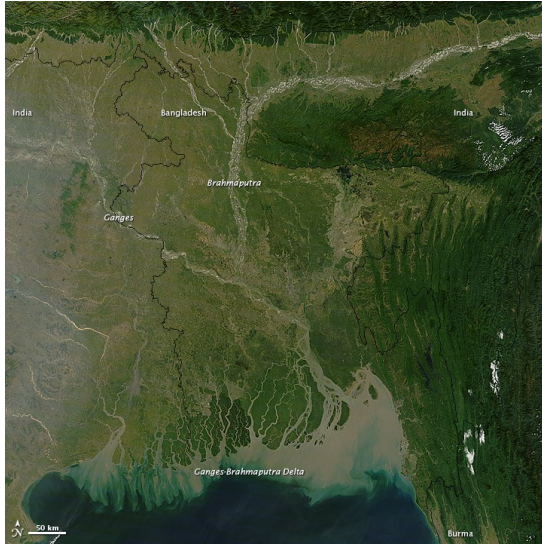


Figure 5: A flexible float using `[!htbp]`.

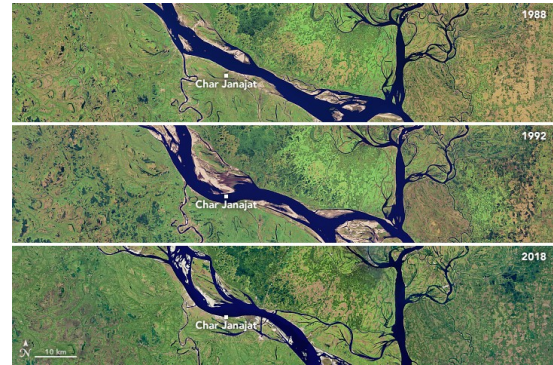
6 Subfigures

Package note: subfigures in this demo need `\usepackage{subcaption}`.

Subfigures need the `subcaption` package. Each image sits inside a `subfigure` block. The block width controls the slot; inside the slot, use `width=\linewidth` so the image fills that slot.



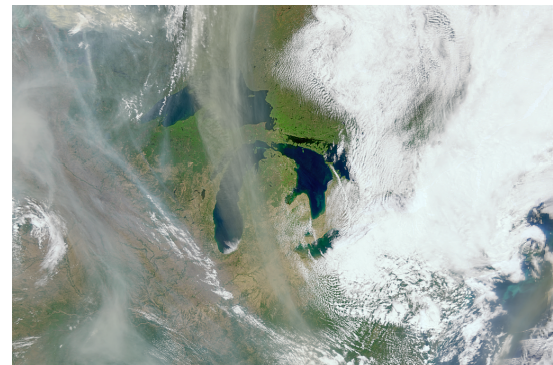
(a) Country scale



(b) River scale



(c) Space scale



(d) Atmospheric view

Figure 7: Four subfigures arranged as a two-by-two grid.

WARNING: DO NOT put blank line between subfigures. It starts a new paragraph, and forces the next subfigure onto a new line, no matter the sizes.

We can reference the full figure as Figure ??, or a single part such as Figure ??. The main layout trick is to make each subfigure slightly less than half the line width. The `\hfill` command separates the two columns, and the blank line with `\vspace` starts the second row.

7 Hyperlinks

Package note: links and clickable references need `\usepackage{hyperref}`.

The `hyperref` package makes cross-references, citations, and web links clickable. Use `\href` when the displayed text should be different from the URL, and `\url` when the URL itself should be printed.

Example: [Overleaf Learn LaTeX](https://www.overleaf.com). A raw URL can also be printed with `\url`: `\url{https://www.overleaf.com}`.

8 Footnotes

Package note: ordinary footnotes are built in; no extra package is needed.

Use `\footnote{...}` for short side comments, source notes, or small clarifications that should not interrupt the main paragraph.¹ Footnotes are numbered automatically, and LaTeX handles their placement.

This claim needs a short explanation.`\footnote{Write the note here.}`

If the same note must be referenced twice, one simple approach is to store the footnote number and reuse it:

First mention.`\footnote{\label{fn:tool}A repeated note.}`
Second mention.`\footnotemark[\ref{fn:tool}]`

Avoid long footnotes in short reports. If the note becomes a full paragraph, it probably belongs in the main text.

9 Paper Citations and Bibliography

Package note: this compiled demo uses BibTeX together with the `natbib` package. Plain BibTeX can work without a citation package, but `natbib` adds better citation commands while still using a `.bib` file and BibTeX.

For CSE reports, students must know how to cite papers, not just websites. The workflow is: keep source details in a `.bib` database, cite those keys in the text, and print the bibliography at the end of the document.

9.1 Plain BibTeX Starting Point

Plain BibTeX begins with the built-in `\cite{...}` command. These lines are intentionally shown as comments because the compiled demo below uses the more expressive `natbib` commands instead.

```
% Single source with plain BibTeX:  
% \cite{vaswani2017attention}
```

```
% Multiple sources with plain BibTeX:  
% \cite{krizhevsky2012imagenet,he2016deep}
```

9.2 Natbib Commands in Action

Use `\citet` when the author name is part of the sentence.

`\citet{vaswani2017attention}` introduced the Transformer architecture.

¹This is a regular footnote. It appears at the bottom of the page.

Compiled result: [?] introduced the Transformer architecture.

Use `\citep` when the citation is parenthetical evidence at the end of a claim.

Residual learning became a standard CNN technique `\citep{he2016deep}`.

Compiled result: residual learning became a standard CNN technique [?].

Multiple papers can still be cited together by putting several keys inside the same command.

`\citep{krizhevsky2012imagenet,he2016deep}`

Compiled result: image classification and residual networks can be cited together [? ?].

Sometimes only the author name or only the year is needed.

`\citeauthor{he2016deep}` proposed residual learning in `\citeyear{he2016deep}`.

Compiled result: [?] proposed residual learning in [?].

The package option controls the broad citation style. This document uses the numeric option:

`\usepackage[numbers]{natbib}`

For an author-year demo, one can try the author-year option together with a compatible bibliography style such as `plainnat`.

`\usepackage[authoryear,round]{natbib}`

9.3 Including Uncited Sources

When a source should appear in the bibliography even though it is not cited in the paragraph, use `\nocite{...}`. This command does not print anything at the current location; it only asks LaTeX and BibTeX to include that entry in the reference list.

`\nocite{lamport1994latex}`

The command `\nocite{*}` prints every entry from the `.bib` file. That can be useful for a bibliography demo, but it is usually too broad for a finished report.

9.4 Bibliography Files

The source details live in `references.bib`. A paper entry usually contains fields such as `author`, `title`, `booktitle` or `journal`, `year`, and sometimes `pages`, `doi`, or `url`.

```
@inproceedings{vaswani2017attention,
  author = {Ashish Vaswani and others},
  title = {Attention Is All You Need},
  booktitle = {Advances in Neural Information Processing Systems},
  year = {2017}
}
```

9.5 Styles and Compile Order

Change the bibliography style to change how references are printed. This document currently uses:

```
\bibliographystyle{plainnat}
\bibliography{references}
```

The style command selects the formatting style. The bibliography command tells BibTeX to read `references.bib`. Common BibTeX-style options include:

- `plainnat`: natbib-friendly plain numeric or author-year style.
- `plain`: numeric labels, sorted alphabetically by author.
- `abbrv`: like `plain`, but with abbreviated names.
- `unsrt`: numeric labels ordered by first citation.
- `ieeetr`: IEEE-like numeric style, common in engineering.
- `apalike`: author-year-like references for simple reports.

10 Algorithms, Pseudocode, and Code Listings

Package note: pseudocode here uses the `algorithm` and `algpseudocode` packages. Code listings use the `listings` package.

Technical reports often need pseudocode or small code snippets, and Overleaf also treats code listing as a standard topic. Use `listings` for code because it compiles normally. Avoid `minted` in an introductory setting unless shell escape is explicitly allowed.

Algorithm 1 Binary Search

Require: Sorted array A , target value x

Ensure: Index of x , or -1 if not found

```
1:  $low \leftarrow 1, high \leftarrow n$ 
2: while  $low \leq high$  do
3:    $mid \leftarrow \lfloor (low + high)/2 \rfloor$ 
4:   if  $A[mid] = x$  then
5:     return  $mid$ 
6:   else if  $A[mid] < x$  then
7:      $low \leftarrow mid + 1$ 
8:   else
9:      $high \leftarrow mid - 1$ 
10:  end if
11: end while
12: return  $-1$ 
```

Listing 1: A small Python function.

```
def greet_user(name):
    # Create a greeting message using an f-string
    message = f"Hello, {name}! Welcome to Python."

    # Print the greeting message
```



```
print(message)

# Return the message in case it's needed elsewhere
return message

# Example usage
user_name = "Alice"
greet_user(user_name)
```

Algorithm ?? and Listing ?? can be referenced just like figures and tables.

11 Two-Column Documents

Package note: document-wide two-column layout can be requested in the document class option, e.g., `\documentclass[twocolumn]{article}`. The separate demo also uses `graphicx`, `subcaption`, `booktabs`, and `hyperref`. Open and compile `two-column-demo.tex`; it uses a full two-column article with `\documentclass[twocolumn]{article}`, normal figures, and a wide figure spanning both columns with `figure*`.

12 Debugging Checklist for This Lesson

- If a reference shows as ??, compile LaTeX again.
- If a citation shows as ?, check the citation.
- If an image is missing, check the file name, extension, and folder.
- If a float is far away, try `[\!htbp]` before forcing `[H]`.
- If a subfigure overflows, reduce each subfigure width before changing manual spacing.
- If colored table cells fail, load `\usepackage[table]{xcolor}`.